



Review

Cutting edge developments in bone oncology – Highlights from the Cancer and Bone Society 2024 annual meeting

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HIGHLIGHTS

- The Cancer and Bone Society (CABS) serves as a global hub for cancer and bone research, uniting basic scientists and clinicians.
- Sessions showcased molecular and clinical breakthroughs, illustrating the translational applicability to patient care.
- Emerging work revealed the critical role of the bone microenvironment in metastasis and therapeutic resistance.
- Cutting-edge tools were presented that reshape how researchers visualize, model, predict, and treat skeletal involvement in cancer.
- Networking, mentorship, and innovative engagement of Early-Stage Investigators ensured a robust and committed future for the field.

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ABSTRACT

The 2024 Annual Conference of the Cancer and Bone Society (CABS), held jointly with the Bone Research Society at the University of Sheffield, highlighted the Society's enduring mission to advance understanding of cancer–bone interactions through collaboration, mentorship, and scientific excellence. Established from the early Cancer Induced Bone Disease meetings and re-emerging as an independent body in 2016, CABS continues to serve as a global community bridging basic, translational, and clinical research. This year's meeting featured a comprehensive scientific program spanning tumor–bone biology, therapeutic innovation, and microenvironmental regulation. Clinical sessions emphasized the evolution of bone-targeted treatments such as bisphosphonates, denosumab, and radiopharmaceuticals, alongside predictive biomarkers to guide personalized care. Presentations on renal cell carcinoma bone metastases, multiple myeloma, and prostate cancer underscored persistent clinical challenges and opportunities for integration of immunotherapies and precision radiotherapy. Molecular sessions revealed how non-coding RNAs, TGF- β /Gli2 signaling, metabolic dependencies, and stromal mediators—including osteomodulin, IL-1 β , and adiponectin—govern tumor colonization, dormancy, and resistance. Advances in immuno-oncology demonstrated the potential of $\gamma\delta$ CAR-T cells, oncolytic viruses, and immune-metabolic targeting to enhance therapeutic efficacy in the bone microenvironment. Spatial and single-cell profiling provided unprecedented resolution of the metastatic niche, identifying novel osteoblast-derived stromal subsets and neurovascular remodeling as key determinants of disease progression. Complementary preclinical models and computational simulations, including 3D engineered bone marrow constructs and micro-finite element analysis, offered new frameworks to study therapeutic responses and fracture risk. The meeting also celebrated the growing leadership of Early-Stage Investigators (ESIs), whose contributions extended beyond research presentations to creative networking and mentorship events. These efforts reflected CABS' commitment to inclusivity, community building, and sustaining the next generation of leaders in cancer and bone research. Collectively, the 2024 CABS Annual Conference captured a transformative moment in the field—where mechanistic discoveries, advanced modeling, and translational collaboration converge to improve outcomes for patients with skeletal malignancies. As CABS looks ahead to the 2026 meeting in Tampa, Florida, the Society

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remains steadfast in fostering global partnerships and advancing integrative approaches to conquer cancer in bone.

1. Introduction

The Cancer and Bone Society (CABS) has become a cornerstone for scientists and clinicians working to unravel the complex relationship between cancer and the skeletal system [1]. What began as the Cancer-Induced Bone Disease (CIBD) meetings in 1991 has evolved into a thriving international society that fosters innovation, mentorship, and cross-disciplinary dialogue. Officially established in 2004 by pioneers Drs. Greg Mundy, Robert Rubens, Etsuro Ogata, Herbie Fleisch, and Jack Martin, CABS built on this foundation under the leadership of its first chair, Dr. Robert Coleman, to create a global community bridging basic science, translational research, and clinical care.

After a brief period as part of the International Bone and Mineral Society (2011–2015), CABS re-emerged as an independent society in 2016. Since then, it has steadily expanded its reach while remaining true to its mission: to advance understanding of cancer's impact on bone through collaboration, scientific excellence, and mentorship of the next generation.

The 2024 CABS Annual Conference in participation with the Bone Research Society [2] held at the University of Sheffield reflected these values throughout its program. CABS President Dr. Rachelle Johnson and CABS President-Elect Dr. Janet Brown opened this year's joint meeting by warmly welcoming attendees from across the globe. Their remarks set the tone for a conference dedicated to advancing the frontiers of cancer and bone research through international collaboration. Sessions highlighted basic, translational, and clinical research ranging from fundamental mechanisms to therapeutic advances, including novel approaches with non-coding RNA and CAR-T cells as well as leveraging new metabolic and dormancy vulnerabilities.

CABS is also distinguished by its strong support for graduate and early career researchers. At the 2024 meeting, the Early-Stage Investigator (ESI) Committee organized events designed to foster mentorship and networking. These events aimed to provide opportunities for trainees to engage with senior scientists, exchange ideas, and build community. By combining high-quality science with a conference program designed to engage the next generation of scientists, CABS continues to shape the future of cancer and bone research. Its long-standing commitment to collaboration, rigor, and mentorship ensures that the Society will remain a leader in the field for years to come.

2. Clinical management of bone metastasis: A history and where we're headed (Session 1)

The *Opening Session* covered the evolution of treatment strategies for cancers that involve bone, including bone metastases and multiple myeloma. For his immense contributions to the cancer and bone field, Dr. Robert Coleman was awarded the CABS 2024 Distinguished Service Award. As the Keynote Speaker, Dr. Coleman provided an overview of how bone targeted treatment strategies for bone metastases have evolved over time, highlighting the importance of bone-targeting agents and explored emerging research and future strategies aimed at improving patient outcomes. Dr. Coleman emphasized that bone-targeted agents, such as bisphosphonates and denosumab, have transformed clinical care by significantly reducing skeletal morbidity (skeletal-related events, such as fractures, spinal cord compression, and hypercalcemia) and improving quality-of-life in patients with metastatic bone disease. In earlier disease settings, especially in breast cancer, he argued that adjuvant bisphosphonates reduce the risk of bone metastases and breast cancer mortality—but that this benefit is markedly influenced by menopausal status, with postmenopausal women deriving greater benefit. He also noted that while denosumab is highly effective

for preventing fractures and treating bone loss, its evidence for reducing disease recurrence or improving survival is less consistent compared to bisphosphonates in many settings. A key future direction in his view is the development of predictive biomarkers (i.e., tumor expression of MAF) to identify which patients are most likely to benefit from bone-targeted agents, thereby personalizing therapy and avoiding over-treatment. He also identifies promise in novel therapeutic strategies beyond traditional antiresorptive drugs, including radiopharmaceuticals (i.e., radium-223), and potentially anabolic agents or approaches that modify the bone microenvironment—to further improve outcomes both in prevention and treatment of bone metastases.

Dr. Christopher Parrish (Leeds Teaching Hospitals NHS Trust, Leeds, UK) provided an updated overview of recent advances in immunotherapy and novel drugs in the treatment of multiple myeloma. The presentation focused on emerging immunotherapies and drug strategies and how these developments are reshaping therapeutic approaches to improve outcomes in patients. Dr. Hanna Taipaleenmäki (Ludwig-Maximilians-University, Munich, GE) discussed the broader impact of anti-resorptive therapies on the musculoskeletal system in preclinical studies citing the association of low muscle radiodensity with decreased survival in breast cancer patients. She discussed important pre-clinical studies on how increased fat deposition within skeletal muscle impacts bone metastasis, focusing on recent evidence supporting the use of anti-resorptives to reverse myosteosis.

3. Emerging clinical breakthroughs (Session 2)

The *Plenary Symposium* on the clinical management of bone metastases defined important clinical challenges in treating specific malignancies impacting bone and presented emerging therapeutic strategies. Dr. Janet Brown (University of Sheffield, Sheffield, UK) reviewed the clinical features and therapeutic challenges associated with renal cell carcinoma bone metastasis, emphasizing the frequency of osteolytic lesions and poor patient outcomes associated with metastatic bone involvement. She also provided updates on recent treatment advancements, including the use of drug combinations and immunotherapies. Expanding the discussion to targeted local treatments, Dr. Peter Dickinson (Leeds Teaching Hospitals NHS Trust, Leeds, UK) discussed the use of stereotactic radiotherapy for bone metastatic lesions—a technique that allows precise administration of ablative radiation doses to patients, resulting in better pain control and local tumor responses compared to conventional radiation therapy. Shifting the focus to primary bone malignancies, Dr. Robin Young (Sheffield Teaching Hospital, Sheffield, UK) stressed the need for more clinical research on osteosarcoma, citing limited treatment progress and challenges posed by the disease's rarity and heterogeneity. He called for identification of new molecular markers to enable personalized treatment and improve patient prognosis. While significant advancements have made progress in treating both primary and secondary bone cancers, significant clinical gaps persist, reinforcing the urgent need for continued innovation and collaboration.

4. Molecular advances in the bone metastatic niche (Session 3)

Key molecular drivers of tumor progression in bone were highlighted in this session for their roles in signaling as influencing bone metastatic risk, bone colonization, and chemoresistance. Opening the session, Dr. Phillipe Clézardin (Inserm, Paris, FR) demonstrated that small non-coding miRNAs can help predict bone metastasis risk in patients. In particular, the presence of miR-662 correlates with bone metastasis in breast cancer patients and promotes breast cancer cell stemness and

migration. These findings underscore the potential for miRNA monitoring and targeted therapy development to help predict and prevent bone metastasis. Addressing mechanisms of treatment resistance, Outstanding Abstract Award winner Ms. Jade Miller (Vanderbilt University, Nashville, TN, US) showed that TGF- β and non-canonical Gli2 activation drives chemoresistance in bone-tropic breast cancer cells and enhanced osteolysis. Importantly, Gli2 blockade improved the efficacy of paclitaxel, supporting the development of Gli2-targeted combination therapies. The theme of microenvironmental drivers of bone metastasis continued with Dr. Robert Clarke (University of Manchester, Manchester, UK) who identified both IL-1 β and osteomodulin (OMD) as key microenvironmental signals promoting breast cancer colonization in bone. Proteomic analysis linked OMD to migratory signaling pathways, suggesting its role in facilitating metastatic spread to the bone niche. Shifting from breast cancer to multiple myeloma, Dr. Young-Eun Park (University of Oxford, Oxford, UK), winner of an Outstanding Abstract Award, revealed that bone marrow-derived adipocytes secrete adiponectin, which drives proliferation and reactivation of dormant myeloma cells. This discovery points to a novel microenvironmental mechanism underlying myeloma progression, opening new avenues for therapeutic intervention. Collectively, this session underscored the complexity of the bone microenvironment and highlighted emerging clinical insights and therapeutic strategies that hold promise for improving patient outcomes.

5. Clinical insights and lightning talks (Session 4)

The *Lightning Talks* session featured rapid five-minute presentations from High Scoring Abstract Award winners and provided a lively showcase of innovative science across the spectrum of cancer and bone research.

Presentations highlighted novel therapeutic strategies and disease mechanisms. Dr. Jacky Wong (University of Sheffield, Sheffield, UK), demonstrated that the bone anabolic agent losartan, responsible for increasing the diameter of blood vessels, mitigates bone loss in breast cancer-induced bone disease *in vivo*. Margherita Puppo (INSERM, Lyon, FR) described the dual inhibitory role of miR-24-2-5p during early breast cancer bone metastasis formation. Dr. Robert Satcher (University of Texas – MD Anderson Cancer Center, Houston, TX, US) revealed how TGF β i drives osteocyte apoptosis and gap junction failure in renal cell carcinoma bone metastasis. Dr. Diane Lefley (University of Sheffield, Sheffield, UK) reported that combining ATM inhibitors with radium-223 reduces prostate cancer bone metastases *in vivo*.

Several talks focused on the bone microenvironment and disease progression. Mr. Karl Nyman (Moffitt Cancer Center, Tampa, FL, US) examined the reciprocal effects of aging stroma on multiple myeloma, and Dr. Gabriel Pagnotti (University of Texas – MD Anderson Cancer Center, Houston, TX, US) presented low-intensity vibration as a means to partially alleviate bone loss and muscle weakness in a long-term assessment of myeloma-bearing mice treated with proteasome inhibitors. Dr. Alexandria Sprules (University of Sheffield, Sheffield, UK) introduced a 3D model of myeloma dormancy for preclinical drug screening.

Metabolic and molecular regulators also took the spotlight. Dr. Renee Ormsby (University of Oxford, Oxford, UK) discussed targeting G6PD in the prostate cancer bone niche, while Mr. Jeremy Kane (Vanderbilt University, Nashville, TN, US) presented on transcriptional regulation of the breast tumor suppressor LIFR by the carboxy-terminal PTHrP domain. Dr. Steven Wood (University of Sheffield, Sheffield, UK) profiled prostate cancer-derived exosomes as circulating mediators and potential biomarkers of metastatic spread to bone. Dr. Emmanuel Massy (Hospices Civils de Lyon, Lyon, FR) reported synergistic effects of denosumab and immune checkpoint inhibitors in metastatic bone disease, and Dr. Ning Wang (University of Leicester, Leicester, UK) identified the glycosylation enzyme ST6GAL1 as a novel target in prostate cancer bone metastasis.

Together, these brief but impactful presentations underscored the diversity and translational potential of ongoing research in the field, spanning novel therapeutic approaches, tumor–bone interactions, and innovative preclinical models.

6. Updates on assessing therapeutic responses in bone metastasis and fracture risk (Session 5)

The following session switched the discussion to highlight advances in assessing and managing fracture risks in cancer patients with skeletal lesions. Strategies to predict and combat skeletal-related adverse events were presented including early intervention of endocrine adjuvant therapies, targeting tumor–bone microenvironmental interactions to enhance fracture repair, and use of computational models to predict vertebral failure.

Dr. Peyman Hadji (Frankfurt Center for Bone Health and Endocrinology, Frankfurt, GE) explored how adjuvant endocrine therapies accelerate bone loss and increase fracture risk and the demographic of patients most at-risk. Emphasis was made on the importance of early risk assessment and bone-protective strategies to prevent long-term skeletal complications as a key to improving outcomes. Dr. Francis Lee (Yale School of Medicine, New Haven, CT, US) examined how cancer cells interact with the bone microenvironment in clinically relevant murine models and in human patients to influence disease progression and treatment outcomes. Therapeutic strategies to target microenvironmental interactions to enhance fracture repair were discussed, including use of the AORIF technique in patients that combines tumor targeting with bone stabilization. To round out the session, Dr. Enrico Dall'Ara (Insigneo Institute, University of Sheffield, Sheffield, UK) discussed the development of computational models to predict how vertebrae with metastatic lesions, dictated by size, type, and location of tumor lesions, fail under load. Specifically, micro-finite element (FE) modeling is used to predict fracture potential in vertebral bone metastases, while reconstructed bone built from quantitative-CT (computed tomography) or preclinical micro-CT imaging is used to predict how bone metastatic lesions affect the strength, deformation, and fracture risk of vertebrae and femora, validated against experimental data. These models show promise for improving fracture risk prediction and could inform better clinical decision tools.

7. Immuno-oncology and the bone microenvironment (Session 6)

The session explored recent advances in understanding immune-tumor interactions within the skeletal niche. A major focus was on the expanding role of immunotherapy in the treatment of bone-associated cancers, alongside the influence of metabolic and inflammatory factors on disease progression.

Dr. Michelle Lawson (University of Sheffield, Sheffield, UK) presented preclinical data on an engineered oncolytic adenovirus driven by the SLAMF7 promoter, demonstrating anti-myeloma activity. This novel therapeutic approach induced caspase-independent cell death and boosted immune responses, reducing myeloma burden and offering potential as a targeted therapy for the disease. In an adoptive cell therapy approach, Dr. Conor Lynch (Moffitt Cancer Center, Tampa, FL, US) introduced a strategy using $\gamma\delta$ CAR-T cells engineered to target the prostate stem cell antigen (PSCA) in bone metastatic castrate-resistant prostate cancer. These cells showed potent cytotoxicity against bone metastases, with enhanced efficacy when combined with zoledronate, highlighting their translational potential. Outstanding Abstract Award winner Dr. Nicola Giuliani (University of Parma, Parma, IT) discussed how metabolic adaptations in myeloma cells contribute to disease progression. Her work revealed that myeloma cells increase glutaminolysis, resulting in glutamate accumulation that promotes osteoclast activation, in part through upregulation of the glutamate transporter SLC1A3. Inhibiting this transporter impaired osteoclast differentiation and

reduced bone resorption in models of myeloma. Addressing microenvironmental crosstalk in the bone marrow niche, Dr. Jayna Mistry (The Jackson Laboratory, Bar Harbor, ME, US), also a winner of the Outstanding Abstract Award, demonstrated that Dnmt3a-mutant hematopoietic cells induce senescence in mesenchymal stromal cells (MSCs), fostering a bone marrow environment conducive to clonal hematopoiesis and malignant progression. Treatment with senolytic agents reversed these effects, supporting the therapeutic targeting of senescent MSCs in hematologic malignancies. These studies underscore the multifaceted role of the bone microenvironment in cancer progression and demonstrate how targeting immune, metabolic, and stromal compartments can lead to new opportunities for therapeutic intervention.

8. Spatial profiling of the bone metastatic niche (Session 7)

Continuing the microenvironment theme, this session explored the intricate spatial and cellular architecture of the bone metastatic niche using advanced imaging, engineered models, and single-cell approaches. Focus was placed on understanding how tumor cells exploit bone-specific microenvironments and how remodeling of the nerve and bone stroma contributes to disease progression.

The session opened with Dr. Ralph Müller (ETH, Zürich, CH) presenting on cutting-edge spatial imaging techniques for visualizing both bone metastatic lesions and bone regeneration *in vivo*. His work demonstrated how 3D visualization of the metastatic niche captures dynamic interactions among tumor cells, the bone matrix, and surrounding stromal components—showing that these tools are helping to unravel site-specific remodeling events that dictate metastatic colonization and progression. Dr. Abdullah Khan (University of Oxford, Oxford, UK) followed with the introduction of an engineered bone marrow model that mimics human bone marrow architecture. This system enables high-resolution investigation of myeloma-microenvironment interactions and supports the tracking of dormancy, proliferation, and therapeutic responses in multiple myeloma within a spatially organized, physiologically relevant context.

Employing single-cell transcriptomics, Outstanding Abstract Award winner Dr. Jennifer Zarrer (Ludwig-Maximilians-University, Munich, GE) identified a previously unrecognized osteoblast-derived cell population within the bone metastatic niche. These cells exhibit fibroblast-like properties and appear to play a crucial role in promoting breast cancer cell survival and outgrowth in bone, revealing a potential new stromal target for therapeutic intervention. Offering new insights into the neurological impacts of the bone niche, Dr. Marta Diaz del Castillo (University of Aarhus, Aarhus, DK) explored compartment-specific remodeling of nerve fibers within the bone marrow during myeloma progression. Her findings revealed that heterogeneous remodeling in the neurovascular niche may influence tumor distribution, pain, and response to neuro-targeted therapies, thus underscoring the importance of spatially resolving nerve-tumor interactions in the bone microenvironment. This session illuminated the critical value of spatially informed approaches in unravelling the complexity of the bone microenvironmental niche and advancing the development of more precise, effective therapies for bone-involved cancers.

9. Plenary oral presentation highlights (Session 8)

The *Plenary Oral Sessions* featured speakers with the top-scoring abstracts. These sessions highlighted innovative and impactful research that spanned fundamental biology, therapeutic resistance, and new treatment approaches for cancer in bone.

Session I opened with Dr. Karina Geurtzen (Katholieke Universiteit, Leuven, BE) demonstrating that breast cancer cells rely heavily on mitochondrial respiration to establish bone metastases, underscoring the metabolic vulnerabilities of tumor cells in this microenvironment. Dr. Beatriz Gamez (University of Oxford, UK) then revealed that high cholesterol levels drive resistance to bortezomib in multiple myeloma,

leading to greater tumor burden *in vivo*, highlighting a potential metabolic-therapeutic interaction with clinical relevance. Ms. Gwenyth Joseph (Vanderbilt University, Nashville, TN, US) concluded the early session by showing that PD-1 blockade, while effective against tumors, can induce musculoskeletal toxicities in a breast cancer bone metastasis model, raising important questions about balancing efficacy with adverse effects.

Session II shifted focus toward tumor-bone interactions and therapeutic targeting. Dr. Ryan Bishop (Moffitt Cancer Center, Tampa, FL, US) presented spatial transcriptomic profiling that mapped the immune landscape of MM, offering new insight into microenvironmental heterogeneity. Dr. Xinhai Wan (University of Texas – MD Anderson Cancer Center, Houston, TX, US) showed that dovitinib, an FDA-approved tyrosine kinase inhibitor, mitigates prostate cancer bone metastases by inhibiting the FGF pathway. Dr. Rebecca Andrews (University of Sheffield, Sheffield, UK) described structural changes in osteocyte lacunae and canalicular networks in a preclinical model, while Dr. Jiabao Zhou (University of Sheffield, Sheffield, UK) reported that combining anakinra with paclitaxel and zoledronic acid reduces metastatic outgrowth of triple-negative breast cancer cells in bone and soft tissues.

Session III highlighted emerging technologies and therapeutic strategies. Dr. James Smith (Garvan Institute, Sydney, AU) used single-cell RNA sequencing to identify conserved components of the dormant tumor cell niche across bone and other metastatic sites, providing a valuable resource for understanding dormancy biology. Ms. Hayley Sabol (University of Arkansas School for Medicine, Little Rock, AK, US) presented a dual-action bone-targeted therapy designed to combat multiple myeloma by eradicating dormant cells while simultaneously improving bone health. Closing the session, Dr. Kaylee O'Donnell (Washington University, St. Louis, MO, US) showed that cAMP-mediated acid sensing by tumor cells enhances immunosuppressive cytokine production and drives tumor progression, pointing to a novel mechanism of immune evasion.

Together, these plenary sessions showcased the breadth of current research in the field, from metabolic dependencies and therapeutic resistance to innovative strategies for targeting tumor-bone interactions and dormancy. They emphasized the translational relevance of mechanistic discoveries and the ongoing effort to balance effective treatment with preservation of skeletal health.

10. Dormancy, senescence, and inflammatory microenvironments (Session 9)

The final plenary session examined the dynamic and often underappreciated roles of dormancy, senescence, and inflammation in shaping the bone marrow microenvironment and influencing cancer progression. Speakers explored how cancer cells exploit these biological states to evade therapy and persist long-term, as well as how stromal and immune components can be co-opted to support metastatic survival or reactivation. Highlighting an unintended consequence of cancer treatment, Dr. Megan Weivoda (Mayo Clinic, Rochester, MN, US) described how anti-cancer therapies can induce senescence in both tumor and stromal cells, triggering the secretion of pro-inflammatory factors that reshape the immune landscape. Her findings showed that senescent cells drive myeloid plasticity, potentially fostering a tumor-permissive environment and contributing to relapse. Dr. Alanna Green (University of Sheffield, Sheffield, UK) addressed the therapeutic challenge posed by dormant myeloma cells, which evade conventional treatments and may later reignite disease. She discussed mechanisms regulating dormancy within the bone niche and outlined strategies to either awaken or eradicate these elusive cells, aiming for more durable responses. Providing mechanistic insight into inflammation-driven remodeling, Dr. Tom Cupedo (Erasmus MC, Rotterdam, NL) explored how chronic inflammation arises from aberrant interactions between stromal and immune cells within the bone marrow. He emphasized that this interplay not only promotes tumor progression but also compromises normal

hematopoiesis and immune surveillance, highlighting the importance of targeting inflammatory circuits as part of cancer therapy. Altogether, the session underscored the complex interactions between cancer cells and the bone microenvironment and pointed to new opportunities for disrupting this interplay to prevent disease recurrence.

11. Early-Stage Investigator-led events

A defining feature of the 2024 Cancer and Bone Society meeting was the strong presence of the Early-Stage Investigator (ESI) Committee, which not only contributed cutting-edge science but also created innovative spaces for career development and community building. These ESI-led events were woven throughout the program and proved to be some of the most engaging aspects of the meeting.

The activities commenced with an interactive *ESI Corner* networking discussion built around the question “*How should we target cancer in bone?*”. This session encouraged junior and senior scientists alike to share perspectives on therapeutic strategies, experimental models, and translational challenges, creating a dynamic exchange that set the tone for the rest of the meeting.

The ESIs hosted the first-ever *Walk and Talk* event, where attendees could break away from traditional on-site interactions and instead connect through conversations while walking, which fostered both scientific exchange and personal connections. Many participants noted that this format made it easier to speak with senior investigators in an approachable, informal setting.

Another *ESI Corner* discussion tackled the theme of “*Alternative Career Paths*.” This was particularly valuable for trainees and post-doctoral fellows considering opportunities outside academia. Senior members with industry, clinical, and policy experience shared candid insights, while early-career attendees were encouraged to ask practical questions about navigating transitions and broadening their professional horizons.

Networking Trivia was a lively event that combined scientific content with social interaction. Trivia teams were intentionally mixed across career stages and institutions, helping break down barriers and encourage collaboration. The lighthearted competition created an atmosphere where junior investigators could comfortably engage with established leaders while also bonding with their peers.

Finally, the ESIs organized *Speed Networking*, a structured but fast-paced event where participants rotated through short conversations with multiple colleagues. This format gave early-career researchers the chance to quickly expand their professional circles, exchange research interests, and plant the seeds for potential collaborations. It also provided a platform for meeting senior faculty and industry partners in an efficient, yet informal way.

Together, these ESI-led events added a unique dimension to the

conference. They complemented the scientific sessions by ensuring that early-career voices were heard and supported, while also strengthening the sense of community that defines the Cancer and Bone Society. The combination of scientific dialogue, career-focused discussions, and interactive networking fostered an inclusive environment that will continue to benefit attendees well beyond the meeting.

12. Concluding remarks

Overall, the 2024 Cancer and Bone Society’s Annual Conference emphasized the latest breakthroughs in cancer and bone research and identified key areas essential for future progress in the field. By bridging gaps among basic, translational, and clinical research while also cultivating the next generation of investigators, the meeting successfully fostered a truly integrated approach to improving outcomes in bone metastasis and primary bone tumors. Building on this momentum, the field now looks ahead to continued collaboration, mentorship, and progress at the 2026 Cancer and Bone Society meeting in Tampa, Florida.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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- [2] www.boneresearchsociety.org.